Regression Theory: Residual Analysis, Diagnostics, Heteroskedasticity and GLS

Lecture Notes

Tuesday, October 13, 2026 & Wednesday, October 14, 2026

1 Residuals

$$\begin{aligned} \underset{\sim}{e} &= \big(I - P(X)\big)\underset{\sim}{Y} \\ E(\underset{\sim}{e}) &= \underset{\sim}{0} \\ \Rightarrow \quad \mathrm{Var}(\underset{\sim}{e}) &= \sigma^2\big(I - P(X)\big) \quad \Longrightarrow \quad \mathrm{Var}(e_i) = \sigma^2(1 - p_{ii}) \\ \mathrm{Cov}\big(\underset{\sim}{e}, \underset{\sim}{\hat{Y}}\big) &= 0 \quad \text{where } \underset{\sim}{\hat{Y}} = P(X)\underset{\sim}{Y} \end{aligned}$$

2 Standardized / Internally Studentized Residuals

$$r_i = \frac{e_i}{\hat{\sigma}_{UE}\sqrt{1 - p_{ii}}}$$

$$e_i \sim N\big(0, \sigma^2(1 - p_{ii})\big) \;\Rightarrow\; \frac{e_i}{\sigma\sqrt{1 - p_{ii}}} \sim N(0,1)$$

$$\hat{\sigma}^2_{UE} = \frac{RSS}{n - p - 1}, \qquad \frac{RSS}{\sigma^2} \sim \chi^2_{n-p-1}$$

Not independent.

Result

$$\frac{r_i^2}{n - p - 1} \;\sim\; \mathrm{Beta}\left(\frac{1}{2}, \frac{n - p - 2}{2}\right)$$

Proof

$$RSS = \underset{\sim}{Y}^T\big(I - P(X)\big)\underset{\sim}{Y}, \qquad \underset{\sim}{e} = \big(I - P(X)\big)\underset{\sim}{Y}.$$

$Q(X) = I - P(X)$ is a projection matrix with rank $n - p - 1$.

$$Q(X) = [\, q_1 \;\cdots\; q_n \,]$$

$$Q(X)\, q_1 = q_1 \quad \text{since } q_1 \in \mathcal{C}(Q(X)), \text{ and } \|q_1\|^2 = 1 - p_{11}.$$

$$\Rightarrow \quad \frac{q_1}{\sqrt{1 - p_{11}}} \text{ is a normalized eigenvector corresponding to eigenvalue 1 of } Q(X).$$

$$Q^T(X) = Q(X), \qquad Q^2(X) = Q(X)$$

We expand $\dfrac{q_1}{\sqrt{1-p_{11}}}$ to $\left\{\dfrac{q_1}{\sqrt{1-p_{11}}}, q_2^*, \ldots, q_{n-p-1}^*\right\}$, an orthonormal basis of $\mathcal{C}(Q(X))$.
$\Rightarrow$ Hence we can have a spectral decomposition of $Q(X)$ as:

$$Q(X) = \frac{q_1 q_1^T}{1-p_{11}} + q_2^* q_2^{*T} + \cdots + q_{n-p-1}^* q_{n-p-1}^{*T}$$

$$A = UDU^T = [u_1 \cdots u_n] \begin{pmatrix} d_1 & & 0 \\ & \ddots & \\ 0 & & d_n \end{pmatrix} \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix} = d_1 u_1 u_1^T + \cdots + d_n u_n u_n^T$$

d_j: eigenvalues of A $\qquad$ u_j: corresponding eigenvector of A.

$$Q(X) = \frac{1}{1-p_{11}} q_1 q_1^T + q_2^* q_2^{*T} + \cdots + q_{n-p-1}^* q_{n-p-1}^{*T} \qquad (*)$$

$$e_1 = \text{1st component of } Q(X)\underset{\sim}{Y} = Q(X)\underset{\sim}{e} = \begin{bmatrix} q_1^T \\ \vdots \\ q_n^T \end{bmatrix} \underset{\sim}{e} = \begin{bmatrix} q_1^T \underset{\sim}{e} \\ \vdots \\ q_n^T \underset{\sim}{e} \end{bmatrix} \qquad \text{as } Q^T(X) = Q(X).$$

$$e_1 = q_1^T \underset{\sim}{e}$$

$$RSS = \underset{\sim}{e}^T Q(X) \underset{\sim}{e} = \frac{(\underset{\sim}{e}^T q_1)^2}{1-p_{11}} + (q_2^{*T}\underset{\sim}{e})^2 + \cdots + (q_{n-p-1}^{*T}\underset{\sim}{e})^2 = \frac{e_1^2}{1-p_{11}} + V^2$$

$$e_1 \perp\!\!\!\perp V$$

$$Q^*(X) = q_2^* q_2^{*T} + \cdots + q_{n-p-1}^* q_{n-p-1}^{*T}$$

$$Q(X) = \frac{1}{1-p_{11}} q_1 q_1^T + Q^*(X)$$

$$Q^{*T}(X) = Q^*(X), \quad Q^{*2}(X) = Q^*(X); \quad q_i^{*T} q_j^* = 0, \quad \|q_i^*\|^2 = 1.$$

$$W = Q^*(X)\underset{\sim}{e}, \quad V^2 = W^T W; \quad W \sim N(0, \sigma^2 Q^*(X))$$

$$e_1 \perp\!\!\!\perp V\,? \quad \text{Yes! since} \quad q_1^T Q^*(X) = 0^T$$

$$V^2 \sim \chi^2_{n-p-2}$$

$$r_i = \frac{e_1}{\sigma\sqrt{1-p_{11}}} \sqrt{\frac{(n-p-1)\sigma^2}{RSS}} = \frac{V_1\sqrt{n-p-1}}{\sqrt{V_1^2 + V^2/\sigma^2}}$$

$$V_1 = \frac{e_1}{\sigma\sqrt{1-p_{11}}} \sim N(0,1), \qquad \frac{V^2}{\sigma^2} \sim \chi^2_{n-p-2}, \qquad \text{independent}$$

$$= \sqrt{n-p-1}\,\frac{(\sigma V_1/V)}{\sqrt{\sigma^2 V_1^2/V^2+1}} = \frac{T}{\sqrt{T^2+1}}$$

$$\frac{r_i^2}{n-p-1} = \frac{V_1^2}{V_1^2+V^2/\sigma^2} \sim \text{Beta}\left(\frac{1}{2}, \frac{n-p-2}{2}\right), \qquad V_1^2 \sim \chi_1^2, \quad \frac{V^2}{\sigma^2} \sim \chi^2_{n-p-2}, \quad \text{independent}$$

$$X \sim \chi^2_{m_1}, \quad Y \sim \chi^2_{m_2}, \quad \text{independent} \quad \Longrightarrow \quad \frac{X}{X+Y} \sim \text{Beta}\left(\frac{m_1}{2}, \frac{m_2}{2}\right)$$

$$E(r_i) = 0$$
$$\text{Var}(r_i) = 1 = E(r_i^2) \quad \text{[Check!]}$$

3 (Externally) Studentized Residual

$$t_i = \frac{e_i}{\hat{\sigma}_{UE,(-i)}\sqrt{1-p_{ii}}} = \frac{\sigma V_1}{\sqrt{V^2/(n-p-2)}} \sim t_{n-p-2}$$

$$\hat{\sigma}^2_{UE,(-i)} = \frac{RSS_{(-i)}}{n-p-2}$$

$$RSS_{(-i)} = RSS - \frac{e_i^2}{1-p_{ii}}, \qquad RSS_{(-i)} = RSS - \frac{e_i^2}{1-p_{11}}$$
$$= \sigma V_1^2 + V^2 - \sigma V_1^2 = V^2$$

$$E(t_i) = 0$$
$$\text{Var}(t_i) = \frac{n-p-2}{n-p-4} \approx 1$$

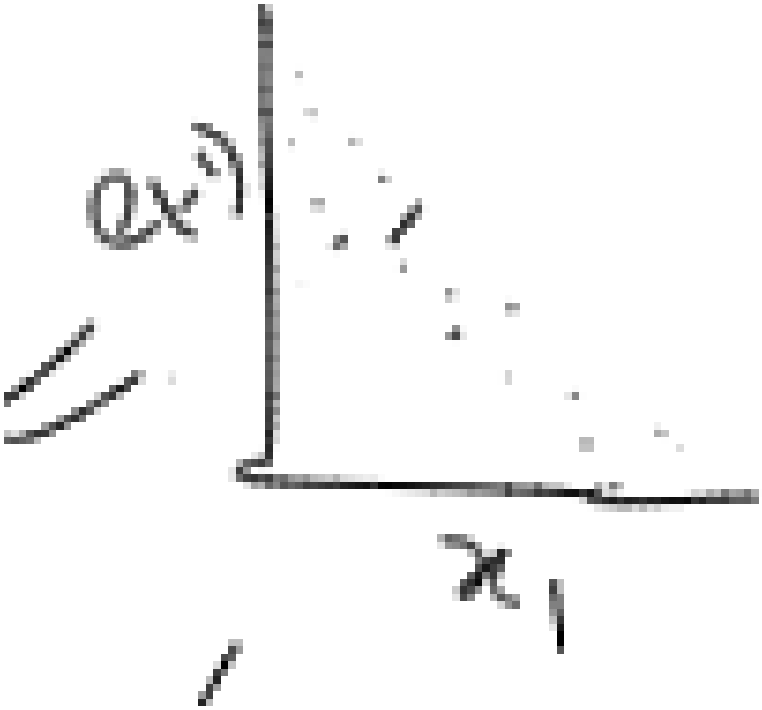
Regression Diagnostics

4 Checking Linearity of the Regressors

(Ref: Fox, Applied Regression Analysis)

$$Y \sim x_2, \ldots, x_p.$$

residual $e_{(1)}$ vs x_1



Also, see the plot of Standardized/Studentized residual vs x_1 (of the model $Y \sim x_1, \ldots, x_p$).

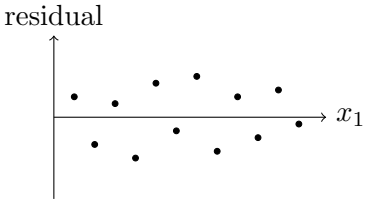


Figure 1: Random scatter: no pattern $\Rightarrow$ linearity assumption holds.

$$\text{Cov}(\underset{\sim}{e}, x_j) = \frac{1}{n}\sum_i (e_i - \bar{e})(x_{ij} - \bar{x}_j) = 0$$

$$Y \sim g(x_1),\, x_2, \ldots, x_p.$$

5 Partial Residual Plot

To identify g.

Suppose (Actual model):

$$Y_i = \beta_0 + \beta_1 g(x_{i1}) + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \varepsilon_i^* \qquad \text{(A)}$$

Working model:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \varepsilon_i, \qquad \varepsilon_i \triangleq \varepsilon_i^* \qquad \text{(W)}$$

Partial residual:

$$e_{i(1)} = e_i + \hat{\beta}_1 x_{i1}, \qquad e_i \text{ obtained by fitting (W)}$$

Plot $(e_{i(1)}, x_{i1})$, $i = 1, \ldots, n$.

Why?

e_i approximates

$$\varepsilon_i = \beta_1 g(x_{i1}) - \beta_1 x_{i1} + \varepsilon_i^*$$
$$\varepsilon_i + \beta_1 x_{1i} = \beta_1 g(x_{1i}) + \varepsilon_i^*$$
$$e_{i(1)} \approx e_i + \beta_1 g(x_{i1})$$

As β_1 does not change with i, plotting $e_{i(1)}$ w.r.t. x_{i1} would reveal the pattern of g.



When does it work?

1) g is indeed linear.
2) x_1 is linearly related to $x_2, \ldots, x_p$.

When does it not work?

1) When g is highly non-linear.

6 CERES Plot

$$\left.\begin{aligned} X_{i2} &= g_2(x_{i1}) \\ X_{i3} &= g_3(x_{i1}) \\ &\vdots \\ X_{ip} &= g_p(x_{i1}) \end{aligned}\right\} \quad \text{approximate } g_2(x_i), \ldots, g_p(x_i)$$

Suppose $\hat{X}_{i2}, \ldots, \hat{X}_{ip}$ are the approximations.

$$Y_i = \alpha + \beta_2 x_{i2} + \cdots + \beta_p x_{ip} + \gamma_{12}\hat{X}_{i2} + \cdots + \gamma_{1p}\hat{X}_{ip} + \varepsilon_i$$

Find the residual from this model, say e_i^*.
Plot e_i^* against x_1.

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7 Partial Residual: Nonlinear Case

Y is not linear in x_1, neither $g(x_1)$. To find g.

Actual model:

$$Y = \beta_0 + \beta_1 g(x_1) + \beta_2 x_2 + \cdots + \beta_p x_p + \varepsilon^* \qquad \text{(A)}$$

Working model:

$$Y = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p + \varepsilon \qquad \text{(W)}$$

Partial residual:

$$e_{i(1)} = e_i + \hat{\beta}_1 x_i \qquad \text{(from (W))}$$

Idea:

$$e \approx \varepsilon = \varepsilon^* + \beta_1 g(x_1) - \beta_1 x_1 \qquad (\hat{\beta}_1 \approx \beta_1)$$

When it works?

1) $g(x_1)$ is linear.
2) x_1 and x_k; $k \neq 1$ are independent (linear).
3) x_1 and x_k are linearly related.

$$x_k = a_k + b_k x_1 + c_k z_k, \qquad z_k \text{ and } x_1 \text{ are linearly independent.}$$

$$(W) \rightarrow Y = \alpha_0 + \alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_p x_p + \varepsilon = \alpha_0^* + \alpha_1^* x_1 + \alpha_2^* z_2 + \cdots + \alpha_p^* z_p + \varepsilon$$

Working model is linear in x_1.

When does PR plot does not work?

- x_k is a non-linear function of x_1 and it is highly related with x_1.

$$x_k = a_k + b_k m_k(x_1) + c_k z_k$$

The working model is now a non-linear function of x_1.
Due to this non-linearity of (W) w.r.t. x_1, the PR plot cannot reveal the form of g.

8 CERES: Conditional Expectation and Residual

Recall, $X_k = g_k(x_1)$, where $X_k = a_k + b_k m_k(x_1) + c_k z_k$, so $E(X_k \mid x_1) \sim m_k$.

Estimate g_k: $\hat{X}_k \sim a_k + b_k m_k(x_1)$.

$\hat{X}_k$ will be much different from X_k (because of z_k).

Also, $\hat{X}_k$ estimates $a_k + b_k m_k(x_1)$, i.e., it estimates the non-linearity present in the working model (W) w.r.t. x_1.

$$(W) \;\to\; Y = \alpha_0 + \alpha_2 x_2 + \cdots + \alpha_p x_p + \gamma_{12}\hat{X}_2 + \cdots + \gamma_{1p}\hat{X}_p + \varepsilon$$
$$e^*_{i(1)} = e^*_i + \hat{\gamma}_{12}\hat{X}_2 + \cdots + \hat{\gamma}_{1p}\hat{X}_p \qquad \text{(obtained from (W))}$$

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9 Heteroskedasticity

$$\mathrm{Var}(\varepsilon) \neq \sigma^2 I$$
$$\Sigma_n = \big((\sigma_{ij})\big), \qquad \frac{n(n+1)}{2} \text{ parameters}$$

Consider the situation: $E(\varepsilon) = 0$; $\mathrm{Var}(\varepsilon) = \mathrm{Diag}(\sigma_1^2, \ldots, \sigma_n^2)$.

Example. Y: expenditure, X: income. $\sigma_i^2 = g(x_i^T \beta)$.

Y: marks in exam. $\sigma_i^2 = \omega(z_i, \lambda)$ [e.g. $\sigma_i^2 = e^{z_i^T \lambda}$]

- X: study time.
- Z: sleep time

Consider two situations: I) $\sigma_i^2 = g(x_i^T \beta)$ II) $\sigma_i^2 = \omega(z_i, \lambda)$.

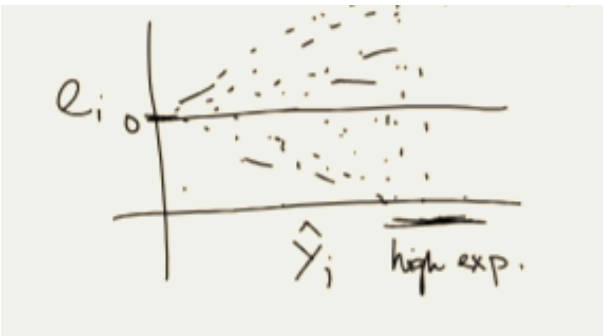
Plots

Example: $\sigma_i^2 = a + b|x_i|$.

$$\underset{\sim}{e} = (I - P(X))\underset{\sim}{Y} = (I - P(X))\varepsilon$$
$$E(\underset{\sim}{e}) = 0, \qquad \mathrm{Var}(\underset{\sim}{e}) = (I - P(X))\Sigma(I - P(X)), \qquad \mathrm{Var}(\varepsilon) = \Sigma = \mathrm{Diag}(\sigma_1^2, \ldots, \sigma_n^2)$$

$$\mathrm{Cov}(\underset{\sim}{\hat{Y}}, \underset{\sim}{e}) = \mathrm{Cov}\big(P(X)Y,\, (I - P(X))Y\big) = P(X)\,\Sigma\,(I - P(X)) \neq 0$$

Fit vs residual plot reveals some pattern. Note $E(\underset{\sim}{\hat{Y}}) = \underset{\sim}{Y}$.



Covariate vs. Residual

$$\mathrm{Var}(\underset{\sim}{e}) = (I - P(X))\Sigma(I - P(X)) = \begin{bmatrix} q_1^T \\ \vdots \\ q_n^T \end{bmatrix} \Sigma [\, q_1 \cdots q_n \,] = \left(\left(q_i^T \Sigma q_j \right) \right)$$

$$\mathrm{Var}(e_i) = (1 - p_{ii})^2 \sigma_i^2 + \sum_{k \neq i} \sigma_k^2 p_{ik}^2$$

(i,i)th component: $q_i^T \Sigma q_i = (-p_{i1}^2 \ \cdots \ -p_{i(i-1)}^2 \ \ (1-p_{ii})^2 \ \ \cdots \ \ -p_{in}^2)(\sigma_1^2, \ldots, \sigma_n^2)^T$.

Recall: $p_{ii} \sim p/n, \qquad p_{ij} \sim 1/n, \qquad n \gg p.$

When n is large, $\mathrm{Var}(e_i) \sim \sigma_i^2 = E(\varepsilon_i^2)$.
If $\sigma_i^2 = \sigma^2 \ \forall i$, then $\mathrm{Var}(e_i) = \sigma^2$.
So, plotting e_i^2, or r_i^2, or t_i^2 against $\hat{Y}_i$ or x_{ik}, $k = 1, \ldots, p$, is helpful in revealing heteroscedasticity.

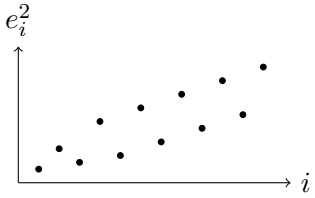


Figure 2: e_i^2 plotted against observation index / covariate – increasing spread indicates heteroskedasticity.

10 Generalized Least Squares (GLS)

$$\Sigma = \mathrm{Diag}\left(\frac{1}{a + bz_1}, \ldots, \frac{1}{a + bz_n} \right)$$

$$\widetilde{Q}_1^2 \sim \chi^2$$

$$\underset{\sim}{Y} = X\beta + \underset{\sim}{\varepsilon}, \qquad \text{where } \underset{\sim}{\varepsilon} \sim N(0, \sigma^2 \Sigma), \ \ \Sigma \text{ known.}$$

$$\Sigma^{-1/2} \underset{\sim}{Y} = \Sigma^{-1/2} X\beta + \Sigma^{-1/2} \underset{\sim}{\varepsilon}$$

$$Y_\Sigma = X_\Sigma \beta + \varepsilon_\Sigma, \qquad \varepsilon_\Sigma \sim N(0, \sigma^2 I) \qquad (*)$$

$$\hat{\beta}_\Sigma = (X_\Sigma^T X_\Sigma)^{-1} X_\Sigma^T Y_\Sigma = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} Y \qquad \text{(GLS solution)}$$

$$\ell(\beta, \sigma^2) = C - \frac{1}{2} \log \det(\sigma^2 \Sigma) - \frac{1}{2\sigma^2} (Y - X\beta)^T \Sigma^{-1} (Y - X\beta)$$

$$\frac{\partial \ell}{\partial \beta} = 0 \qquad \text{and} \qquad \frac{\partial \ell}{\partial \sigma^2} = 0$$

$$\Rightarrow \quad \hat{\beta}_\Sigma = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} Y, \qquad \hat{\sigma}^2 = \frac{1}{n} (Y - X\hat{\beta})^T \Sigma^{-1} (Y - X\hat{\beta})$$

In particular, $\Sigma = \mathrm{Diag}(\sigma_1^2, \ldots, \sigma_n^2)$.

AR(1):

$$\Sigma = \frac{\sigma_\varepsilon^2}{(1-\rho)} \begin{bmatrix} 1 & \rho & \rho^2 & \cdots & \rho^{n-1} \\ \rho & 1 & \rho & \cdots & \rho^{n-2} \\ \vdots & & & \ddots & \vdots \end{bmatrix}$$

1) Observe that $\hat{\beta}_\Sigma$ is also the OLS solution from the model $Y_\Sigma = X_\Sigma \beta + \varepsilon_\Sigma$.

$$\Rightarrow \hat{\beta}_\Sigma \text{ minimizes } \|Y_\Sigma - X_\Sigma \beta\|^2 = (Y - X\beta)^T \Sigma^{-1} (Y - X\beta).$$

2) When $\Sigma = \mathrm{Diag}(\sigma_1^2, \ldots, \sigma_n^2)$, then

$$(Y - X\beta)^T \Sigma^{-1} (Y - X\beta) = \sum_{i=1}^n \frac{1}{\sigma_i^2} (y_i - x_i^T \beta)^2.$$

This is why $\hat{\beta}_\Sigma$ is called the **Weighted Least Squares** solution.

General Case: Σ Unknown

Now consider the general case where $\Sigma = \mathrm{Diag}(\sigma_1^2, \ldots, \sigma_n^2)$, but Σ is not known.
Assume $\sigma_i^2 = \omega(z_i, \lambda)$ and for $\lambda = \lambda_0$, $\sigma_i^2 = \sigma^2$.

Example: $\sigma_i^2 = e^{z_i^T \lambda}$, $\lambda_0 = 0$. ω: known function; z_i's are observed.

We can consider ML estimation:

$$\underset{\sim}{Y} = X\beta + \underset{\sim}{\varepsilon}, \qquad \underset{\sim}{\varepsilon} \sim N(0, \Sigma), \qquad \Sigma = \mathrm{Diag}\big(\omega(z_1, \lambda), \ldots, \omega(z_n, \lambda)\big)$$

$$\ell(\beta, \lambda) = C_0 - \frac{1}{2} \log \det(\Sigma) - \frac{1}{2} (Y - X\beta)^T \Sigma^{-1} (Y - X\beta)$$

$$\frac{\partial \ell(\beta, \lambda)}{\partial \beta} = 0, \qquad \frac{\partial \ell(\beta, \lambda)}{\partial \lambda} = 0, \qquad \sigma_i^2 = \omega(z_i, \lambda)$$

$$\hat{\beta} = (X^T \Sigma^{-1} X)^{-1} X^T \Sigma^{-1} Y \qquad (1)$$

$$\frac{\partial \ell}{\partial \lambda} = \sum_{i=1}^n \frac{\partial \ell}{\partial \sigma_i^2} \cdot \frac{\partial \sigma_i^2}{\partial \lambda} = \sum_{i=1}^n \left\{ -\frac{1}{2\sigma_i^2} + \frac{1}{4\sigma_i^4} (y_i - x_i^T \beta)^2 \right\} \frac{\partial \sigma_i^2}{\partial \lambda} = 0 \qquad (2)$$

$$\frac{\partial \ell}{\partial \sigma_i^2} = \frac{\partial}{\partial \sigma_i^2} \left[c - \frac{1}{2} \log \sigma_i^2 - \frac{1}{2\sigma_i^2} (y_i - x_i^T \beta)^2 \right] = -\frac{1}{2\sigma_i^2} + \frac{1}{4\sigma_i^4} (y_i - x_i^T \beta)^2$$

Example: $\sigma_i^2 = e^{z_i^T \lambda} \quad \Rightarrow \quad \dfrac{\partial \sigma_i^2}{\partial \lambda} = \sigma_i^2 \, z_i.$

$$\beta_0, \lambda_0 \longrightarrow \Sigma \longrightarrow \beta \longrightarrow \lambda \longrightarrow \cdots \qquad \text{(iterative scheme)}$$